

Neonatology: The Micro-Battalion

Precision in the
smallest scale.
Stay fluffy!



Doc FrizzleBob

The System



We turn lists into stories.
Every clinical concept is a character.

The Arena



The Final Frontier of Transition & Immaturity.
We are the External Womb.

How to Read the Deck

1) Character
Name

2) STAT
(Hard diagnostic
data)

3) Visual Prompt
(Neon-Cyberpunk NICU)

4) POWER
(Clinical Move)

5) LORE
(Mechanistic rule-text)



MedKayfabe!

01 **Preemie-Paul**
The Premature

STAT: < 37 weeks gestation
VLBW < 1500g
ELBW < 1000g



POWER: The Transparency-Armor. Paul loses heat and water at record pace due to paper-thin skin.
LORE: Lacks the Brown-Fat-Battery. Brain prone to IVH-the-Bleed (fragile germinal matrix).

02 **Surfactant-the-Slippery**
RDS

STAT: L/S Ratio < 2:1
Ground-glass opacity



POWER: The Anti-Collapse. Reduces surface tension. Prevents the Alveolar-Balloons from snapping shut.
LORE: Prevents Hyaline Membrane Disease. We must spray Liquid-Lung-Shield into his pipes.

03 **Duct-the-Persistent**
PDA

STAT: Machinery murmur
Left-to-Right shunt



POWER: The Lung-Flooder. Fails to close, shunting blood from Aorta-Highway back into lungs.
LORE: Prostaglandins keep it open. Use Indo-the-Lock to shut the gate.

04 **Necro-the-Entero**
NEC

STAT: Pneumatosis intestinalis
(Air in wall)
Abdominal distension



POWER: The Gut-Eroder. Bacterial riot in the immature gut leading to Belly-Burst.
LORE: Triggered by ischemia and Nalve-Gut feeds. Requires Bowel-Rest and Antibiotic-Artillery.

05 **CPAP-the-Breeze**
Airway Splint



STAT: 5-8 cmH2O
Non-invasive
PEEP effect



POWER: The Constant-Open. Splints pipes open so the Knight doesn't fight for every breath.
LORE: Positive pressure prevents alveolar collapse, delaying the Vandal-Ventilator.

06 **Caffeine-the-Jumpstart**
Apnea of Prematurity

STAT: Load 20mg/kg
Maint 5-10mg/kg



POWER: The Breathing-Reminder. Tells the Brain-Stem-Clock to keep ticking.
LORE: Premies have an immature respiratory center. Caffeine increases CO2 sensitivity.

07 **Meconium-the-Clog**
MAS

STAT: Green-stained fluid
Patchy infiltrates



POWER: The Sticky-Invasion. The Knight breathes his first Poop, causing a Chemical-Fire.
LORE: Causes air-trapping and persistent pulmonary hypertension (PPHN).

08 **BPD-the-Scar**
Bronchopulmonary Dysplasia

STAT: Oxygen need at 36 weeks PMA
FiO2 > 21%



POWER: The Chronic-Stiff-Lung. The toll taken when the Vandal-Ventilator damages developing lungs.
LORE: The Alveolar-Balloons become scarred. The heavy price of early survival.

Diagnosis?!

09 **IVH-the-Bleed**
Intraventricular Hemorrhage

STAT: Grade 1-4
Germinal Matrix origin



POWER: The Basement-Flood. Fragile vessels burst, spilling blood into the brain's Water-Rooms.
LORE: Common in ELBW. Grade 3/4 leads to Hydrocephalus-the-Pressure.

10 **HIE-the-Hypoxic**
Encephalopathy

STAT: Sarnat Staging
Low Apgar / Acidosis



POWER: The Oxygen-Blackout. Occurs when the Oxygen-Line is cut during Birth. Brain-Power failure.
LORE: Needs Coc-the-Cap-Therapy to slow cellular death.

11 **PVL-the-White-Matter**
Leukomalacia

STAT: Periventricular
cysts on US / Pre-CP



POWER: The Wiring-Damage. Destroys White-Matter-Cables, destroying future Motor-Moves.
LORE: Caused by pressure drops. The motor wiring literally melts away.

12 **Seizure-the-Spark**
Neonatal Seizures

STAT: Phenobarbital
/ EEG monitoring



POWER: The Electrical-Glitch. Can be silent—Just Lip-Smacking or Bicycling-Legs.
LORE: A surface symptom of underlying HIE or metabolic Stat-Errors.

KayfaBONGO!



13. Bili-the-Yellow

Jaundice

STAT: Total Serum Bilirubin (TSB)
Phototherapy nomogram



POWER: The Skin-Stainer. The Liver-Factory is too slow to process red cell breakdown.
LORE: If Bili crosses the blood-brain barrier: Kernicterus (permanent damage).

14. Hypo-the-Hungry

Hypoglycemia

STAT: Blood glucose < 40 mg/dL
Jitteriness



POWER: The Energy-Crash. Common in infants of diabetic mothers cut off from the High-Sugar-Stream.
LORE: Hyperinsulinism in utero leads to a massive crash when the cord is cut.

15. Gastroschisis-the-Open-Belly

Defect right of cord

STAT: Right of the cord
No sac
High AFP

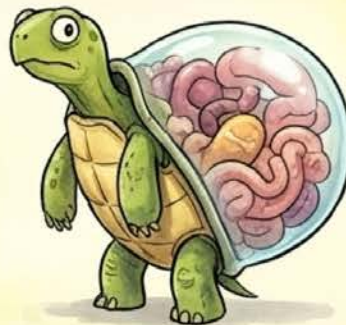


POWER: The Exposed-Gut. Intestines are born Outside the body, naked and angry.
LORE: Requires the Silo-Bag-Squeeze to slowly coax the organs back inside.

16. Omphalocele-the-Sac-Belly

Midline defect

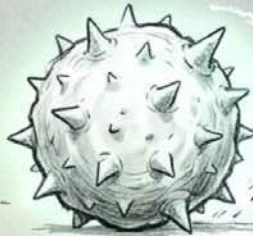
STAT: Midline
Covered by sac
T13, T18



POWER: The Protected-Hernia. Outside the body, but protected inside a Safety-Bag.
LORE: High risk of associated genetic Gaps (Beckwith-Wiedemann).

HUMBUG!





Call Doc Frizzle!

GBS-the-Ghost

Group B Strep



STAT: *S. agalactiae*
Early-onset pneumonia

POWER: The Silent-Sepsis. Hides in the birth canal and ambushes the newborn.
LORE: Needs Intrapartum Antibiotic Prophylaxis (IAP) at weeks 35-37.

TORCH-the-Intrauterine

Infections



STAT: Toxoplasmosis,
Rubella, CMV, Herpes

POWER: The Womb-Sabotage. Attacks the Knight before birth, causing Small-Size and Brain-Scars.
LORE: CMV is the leading cause of non-genetic hearing loss.

Candida-the-White-Patch

Thrush



STAT: Oral Nystatin
Topical Clotrimazole

POWER: The Fungal-Clog. Makes feeding painful and skin raw.
LORE: Invades after Antibiotic-Artillery slaughters the Good-Guard-Bacteria.

Sepsis-the-Systemic-Riot

Systemic infection



STAT: Temp instability
Neutropenia / Lethargy



POWER: The Total-Body-Shutdown. Newborns don't get fevers; they get Quiet-and-Grey.
LORE: High meningitis risk. Deploy the Amp-and-Gent-Guards.



KayfaBOGGLE!

21. The Giraffe-the-Incubator

Omnibed

STAT: Humidification > 80%
for ELBW



POWER: The Artificial-Womb. Keeps the Knight Warm-and-Wet to conserve vital energy.
LORE: Prevents devastating Transepidermal Water Loss (TEWL).

22. The UVC-the-Umbilical-Line

Central access

STAT: Umbilical Venous Catheter
/ Central access



POWER: The Direct-Highway. Immediate central access for High-Concentration-TPN and pressors.
LORE: Must verify the tip placement on X-ray before use.

23. The Neopuff-the-Breather

T-piece resuscitator

STAT: T-piece resuscitator
Controlled PIP



POWER: The Gentle-Inflation. Provides exact pressure without Vandalizing the delicate lungs.
LORE: Unlike manual self-inflating bags, the Neopuff guarantees consistent PEEP.

24. The Phototherapy-Blue-Light

Photoisomerization

STAT: 450-470 nm
Photoisomerization



POWER: The Yellow-Melter. Physically transforms Bili-the-Yellow into water-soluble Lumirubin.
LORE: Protect the Knight's eyes with Neon-Shades!

25. Vitamin-K-the-Clotter

Prophylaxis

STAT: 1mg IM / Week 0



POWER: The Bleed-Blocker. Prevents Vitamin K Deficiency Bleeding (VKDB).
LORE: Newborns possess zero K-Storage. Without it, they bleed into the brain.

26. Erythro-the-Eye-Shield

Ointment

STAT: 0.5% Erythromycin ointment



POWER: The Blindness-Barrier. The first defense against Gonorrheal conjunctivitis.
LORE: A mandatory Delivery Room buff applied to every Knight.

27. Prosta-the-Plug

Alprostadil

STAT: PGE1 infusion



POWER: The Life-Line-Extension. Keeps Duct-the-Persistent OPEN in cyanotic heart defects.
LORE: Buys time and keeps the baby alive until the Surgical-Team arrives.

28. Indo-the-Lock

Indomethacin

STAT: Prostaglandin inhibitor / PDA closure



POWER: The Gate-Closer. Forces the Duct to snap shut, halting the Lung-Flood.
LORE: Also heavily used to prevent IVH in extremely premature babies.

MedKayfabe!

29. PPHN-the-Pressure

Pulmonary Hypertension

STAT: Pre/Post-ductal
SpO2 gap > 10% / R-to-L shunt



POWER: The Valve-Reversal. Lung-pipes clamp shut, forcing blood to completely skip the lungs.
LORE: Devastating high resistance. Requires Nitric-the-Gas to relax the iron grip.

30. Nitric-the-Gas

Inhaled Nitric Oxide

STAT: 20 ppm
Selective vasodilator



POWER: The Pipe-Opener. Selectively relaxes the pipes only in the lungs, not the systemic body.
LORE: Targeted cGMP-activation. The ultimate weapon against PPHN.



31. Hydrocort-the-Stress-Shield

Refractory shock

STAT: Low-dose for
refractory shock



POWER: The Vessel-Sensitizer. Primes the heart and pipes to finally hear adrenaline signals.
LORE: Deployed when the baby's own Adrenal-Factory is too immature to mount a stress response.

32. Dopamine-the-Driver

Inotrope

STAT: 2-20 mcg/kg/min
Inotrope



POWER: The Pump-Boost. Mechanically drives blood pressure up when the Micro-Heart fails.
LORE: The first-line pressor deployed in many NICUs to fight Hypotension-the-Hollow.

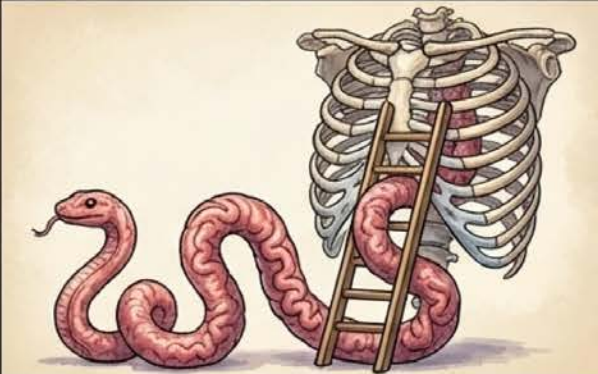
Diagnosis?!



33. CDH-the-Hole

Diaphragmatic Hernia

STAT: Scaphoid abdomen
Bowel sounds in chest



POWER: The Space-Occupier. The gut invades the chest, brutally squashing the Lung-Balloons.
LORE: Usually left-sided (Bochdalek). Lung hypoplasia is the true killer, not the hernia itself.

34. TEF-the-Miswire

Tracheoesophageal Fistula

STAT: Excessive drooling
Choking on feed / Air in stomach



POWER: The Pipe-Miser. The Food-Pipe and Wind-Pipe are dangerously soldered together.
LORE: Associated with VACTERL. Food dumps to lungs; air dumps to stomach.

35. Choanal-the-Closed-Nose

Atresia

STAT: Cyanosis relieved
by crying / NG tube fail



POWER: The Nose-Block. Newborns are obligate nose-breathers. Blocked nose = choking until crying.
LORE: Crying forces open the Mouth-Gate, bypassing the blocked nose temporarily.

36. Hirschsprung-the-Static

Megacolon

STAT: No meconium > 48h
Squirt sign



POWER: The Nerve-Gap. A segment of colon is missing Neural-Drivers, remaining completely Static.
LORE: Aganglionic segment causes everything upstream to balloon into a Mega-Colon.

MedKayfabe!

37 Colostrum-the-Gold

First Milk

STAT: High IgA / Low volume / First 48h



POWER: The Immune-Primer. Paints the Gut-Gate with antibodies.

LORE: The First-Vaccine. The ultimate defense against Necro-the-Entero.

38 HMF-the-Fortifier

Human Milk Fortifier

STAT: Added to breast milk for Premies



POWER: The Growth-Boost. Mimics the Placental-Feed with hyper-dense protein and calcium.

LORE: Premies miss the 3rd-trimester Nutrient-Dump and require massive extra fuel.

39 TPN-the-Vein-Feed

Neonatal

STAT: Glucose Infusion Rate (GIR) / Amino Acids



POWER: The Bypass-Fuel. When the gut is offline, the blood directly carries the food.

LORE: Pure chemical fuel essential for building the brain and bones of ELBW knights.

KayfaBINGO!



40 Osteo-the-Soft-Bone

Metabolic Bone Disease

STAT: High ALP / Low Phosphorus / Premie risk



POWER: The Skeleton-Sag. Without womb-level calcium, the bones turn to soft rubber.

LORE: Osteopenia of Prematurity. Requires aggressive high-dose mineral supplementation.



41. HIE-Cool-the-Cap Therapy

STAT: 33.5°C for 72 hours



POWER: The Chemical-Freeze. Slows metabolic time to halt Excitotoxicity in the brain.
LORE: Must initiate within 6 hours of birth to save the Knight's brain.

42. ROP-the-Laser Treatment

STAT: Anti-VEGF or Laser
photocoagulation



POWER: The Vessel-Burner. Destroys Wild-Vessels before they can peel the retina off.
LORE: A brutal trade: sacrifices peripheral vision to guarantee central vision.

43. IVH-the-Drain Shunt

STAT: Ventriculoperitoneal (VP) Shunt



POWER: The Pressure-Release. Manually vents blood-stained fluid from the brain to the belly.
LORE: The mechanical fix when Hydrocephalus forces the skull to expand too fast.

44. NAS-the-Console Care

STAT: Eat, Sleep, Console (ESC) model



POWER: The Sensory-Shield. Dark rooms and swaddling defeat the chaotic Withdrawal-Riot.
LORE: The first-line behavioral defense to reduce Morphine reliance in drug-exposed babies.

45 The Apgar-Scribe
Assessment

STAT: 1 min & 5 min scores



POWER: The Transition-Judge. Scores whether the Knight is winning the brutal fight to breathe air.
LORE: Heart rate is the single most heavily weighted Stat in the score.

46 The Pre/Post-Ductal-Sat
Screening

STAT: Pulse ox on Right Hand
vs. Either Foot



POWER: The Shunt-Detector. A numerical gap proves blood is skipping the lungs via the Duct.
LORE: The universal Gold-Standard screen for critical congenital heart defects.

47 The Red-Reflex Sentry
Eye Exam

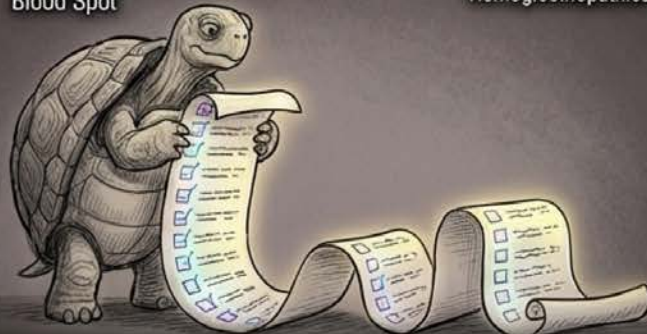
STAT: Ophthalmoscope exam



POWER: The Vision-Guard. If the reflex bounces back white, Retino-the-White-Reflex is hiding inside.
LORE: The simplest, fastest check that universally saves eyes and lives.

48 The Newborn-Screen-Scroll
Blood Spot

STAT: TSH, PKU, Galactosemia,
Hemoglobinopathies



POWER: The Hidden-Glitches-Findar. Sweeps the blood for metabolic errors before symptoms hit.
LORE: Executed precisely at 24-48 hours of life to prevent permanent brain damage.



MedKayfabe!

49. Poly-the-Thick-Blood

Polycythemia

STAT: Hct > 65% / Ruddy complexion



POWER: The Sludge-Flow. Blood becomes so dense it physically jams the Micro-Pipes, triggering seizures or NEC.
LORE: A classic trap for Small-for-Gestational-Age (SGA) knights.

50. Hydrops-the-Total-Flood

Edema

STAT: Edema in >2 compartments (Pleural, Ascites, Skin)



POWER: The Fluid-Overload. The body entirely loses fluid regulation, drowning in its own water.
LORE: Immune (Rh) or Non-immune (Heart/Parvo) triggers. Extremely high mortality.

51. Anemia-of-Prematurity

Low RBC

STAT: Low Retic count / Erythropoietin deficiency



POWER: The Oxygen-Carrier-Shortage. The Marrow-Factory goes on strike, halting red cell production.
LORE: Immature kidneys simply refuse to synthesize enough EPO until the baby ages.

52. Osteo-the-Rickets

NICU Bone Disease

STAT: High ALP / X-ray fraying of metaphysis



POWER: The Fragile-Frame. Depleted calcium/phosphorus causes bones to literally snap during routine Nappy-Changes.
LORE: The devastating result of missing the massive 3rd-trimester placental calcium dump.

What the FLUFF?!



KayfaBINGO!

The Neonatal Attending Doc FrizzleBob

STAT: Master of the Micro-Scale.



POWER: The Threshold-Guardian.

LORE: Respect the immaturity. Titrate the oxygen. Is the heart machine-like? Watch the belly or lose the knight! Stay fluffy!

Synthesis Quest 1: The Premie Flood



Card 01



Card 03



Card 28

LORE: 28-week Paul-the-Premie has a Machinery-Murmur and wet lungs.

QUEST: Identify the Lung-Flooder (Duct-the-Persistent) -> Apply the Gate-Locker (Indo-the-Lock).

Synthesis Quest 2: The Yellow Stainer



Card 13



Card 10



Card 24

LORE: A 48-hour Knight is glowing yellow and unbearably sleepy.

QUEST: Identify the Skin-Stainer (Bili-the-Yellow) -> Apply Blue-Light-Melt (Phototherapy) to evade Kernicterus.

Synthesis Quest 3: The Green Clog



Card 07



Card 30

VICTORY!

LORE: A full-term Knight born through green fluid is gasping. X-ray shows patchy gunk.

QUEST: Identify the Sticky-Invasion (Meconium-the-Clog) -> Anticipate Valve-Reversal (PPHN) -> Prepare Nitric-the-Gas.